What would you like to know about this coin?

H H T H T H T H H H T T H T H

H T H T T H H H T T H T H T H

r is the unknown chance of heads.

> Do we want one guess, or uncertainty about the guess?

One chosen ordered sequence · 17 H and 13 T · recorded in science.json

The data are fixed; the unknown is not a data point

Observed $\mathcal{D}$

$$17 \text{ heads}, \quad 13 \text{ tails}$$

Unknown r

$$0 < r < 1$$

Inference describes plausible values of r.

> Does inferring r change any of the thirty tosses?

A prior is a distribution over the unknown

$$r \sim \mathrm{Beta}(5, 3)$$

$$p(r \mid \mathcal{D}) \;\propto\; p(\mathcal{D} \mid r)\, p(r)$$

> What two ingredients turn it into a posterior shape?

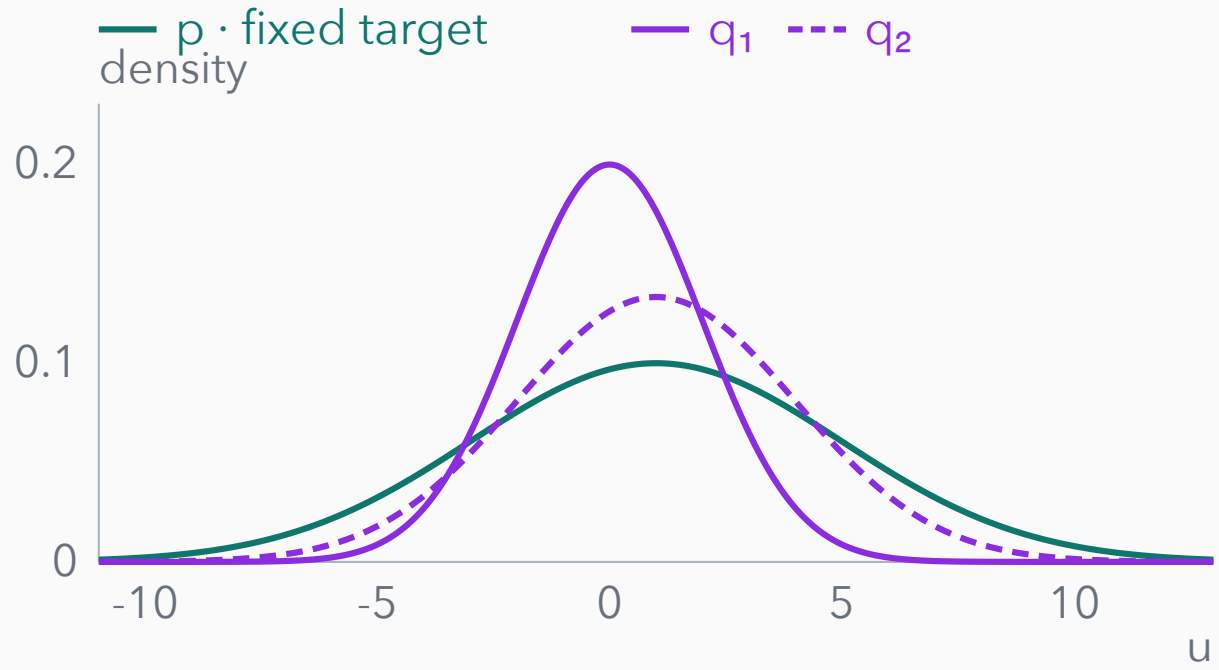
What normalises the posterior?

$$p(r \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid r)\, p(r)}{p(\mathcal{D})}$$

This coin is solvable. We will use its answer to check a more general method.

> Why practise an approximate method on a solvable problem?

Which Gaussian is closer to the target?



$$p = \mathcal{N}(1, 4^2)$$

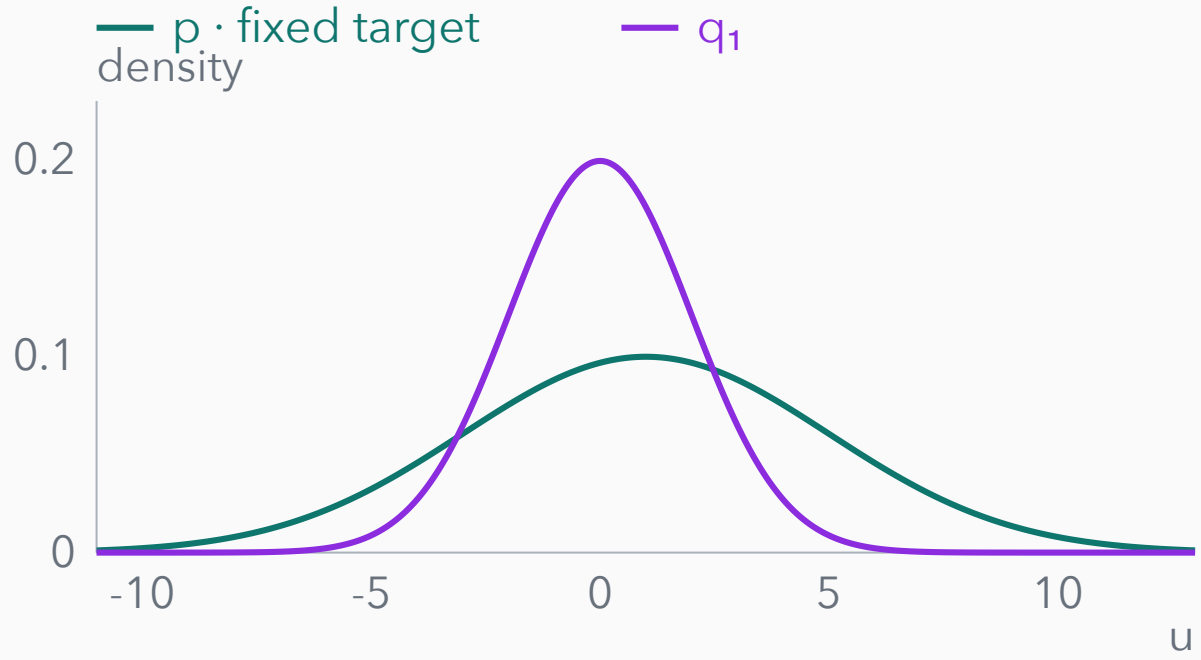
$$q_1 = \mathcal{N}(0, 2^2)$$

$$q_2 = \mathcal{N}(1, 3^2)$$

Normal(loc=μ, scale=σ). The mathematical second argument is variance.

Vote for q_1 or q_2 before calculating.

One density comparison is not the whole story



$$u = 0$$

$$\log q_1(0) - \log p(0)$$

This compares one location. What about the rest of the curve?

> Could this one location settle which whole distribution is closer?

Average the log ratio under q

$$\mathrm{KL}(q\|p) = \mathbb{E}_{u\sim q}[\log q(u) - \log p(u)]$$

The first argument supplies the weighting distribution.

> What does the order q, then p tell us?

Reversing KL changes the question

$$\mathrm{KL}(q_1 \| p)$$

$$\approx 0.3494$$

$$\mathrm{KL}(p \| q_1)$$

$$\approx 0.9319$$

> What does the numerical difference rule out?

Nonnegative KL does not make every term positive

$$\log q(u) - \log p(u) \quad \text{can be negative}$$

$$\mathrm{KL}(q\|p) \geq 0, \qquad \mathrm{KL}(p\|p) = 0$$

Which nonnegative statement applies to the whole average?

For two Gaussians, evaluate the expectation exactly

$$q = \mathcal{N}(\mu, \sigma^2), \qquad p = \mathcal{N}(m, s^2)$$

$$\mathrm{KL}(q\|p) = \log \frac{s}{\sigma} + \frac{\sigma^2 + (\mu - m)^2}{2s^2} - \frac{1}{2}$$

> Where do width and centre mismatch enter?

Substitute first; then compare the score

$$q_1: \quad \log 2 + \frac{4+1}{32} - \frac{1}{2}$$

$$\mathrm{KL}(q_1 \| p) \approx 0.3494$$

$$q_2: \quad \log \frac{4}{3} + \frac{9+0}{32} - \frac{1}{2}$$

$$\mathrm{KL}(q_2 \| p) \approx 0.0689$$

> Does the score support the earlier prediction?

A comparison can become a search

Keep the target fixed. Change q's parameters.

$$\psi^* = \arg\min_{\psi} \mathrm{KL}(q_\psi \| p)$$

> What does this argmin select?

A moment identity explains the squared shift

$$u - m = (u - \mu) + (\mu - m)$$

$$\mathbb{E}_q[(u - m)^2] = \mathbb{E}_q[(u - \mu)^2] + (\mu - m)^2$$

Why does the remaining random term become σ²?

Move the centre while keeping the width fixed

$$\mu = 0, \quad \sigma = 2, \qquad p = \mathcal{N}(1, 4^2)$$

$$L(\mu) = \underbrace{\log 2 + \frac{4}{32} - \frac{1}{2}}_{\text{constant}} + \frac{(\mu - 1)^2}{32}$$

> Which term changes when μ changes?

A negative derivative produces a rightward descent step

$$\frac{\partial L}{\partial \mu} = \frac{2(\mu - 1)}{32}$$

$$= \frac{\mu - 1}{16}$$

$$\left.\frac{\partial L}{\partial \mu}\right|_{\mu=0} = -0.0625$$

Does a negative derivative mean move left under gradient descent?

One literal update

$$\mu^{+} = \mu - \eta \frac{\partial L}{\partial \mu}$$

$$\eta = 4, \qquad \mu^{+} = 0 - 4(-0.0625)$$

$$\mu^{+} = 0.25$$

Which parameter has not changed?

The new curve earns a new loss evaluation



$$\mu : 0 \longrightarrow 0.25$$

KL before	KL after
0.3494	0.3357

σ=2 throughout. Same target and axes.

Did the computed objective improve?

Perfect mean fitting still leaves a gap

$$\mu = 1, \quad \sigma = 2 \neq 4$$

$$\min_{\mu} \mathrm{KL}(q_\mu \| p) \approx 0.3181 > 0$$

> Why is the minimum KL still positive?

A standard deviation must remain positive

$$\rho \in \mathbb{R}, \qquad \sigma = \exp(\rho) > 0$$

```
mu = Parameter(tensor(0.))
rho = Parameter(tensor(log(2.)))
q = Normal(mu, rho.exp())
```

What does ρ represent here?

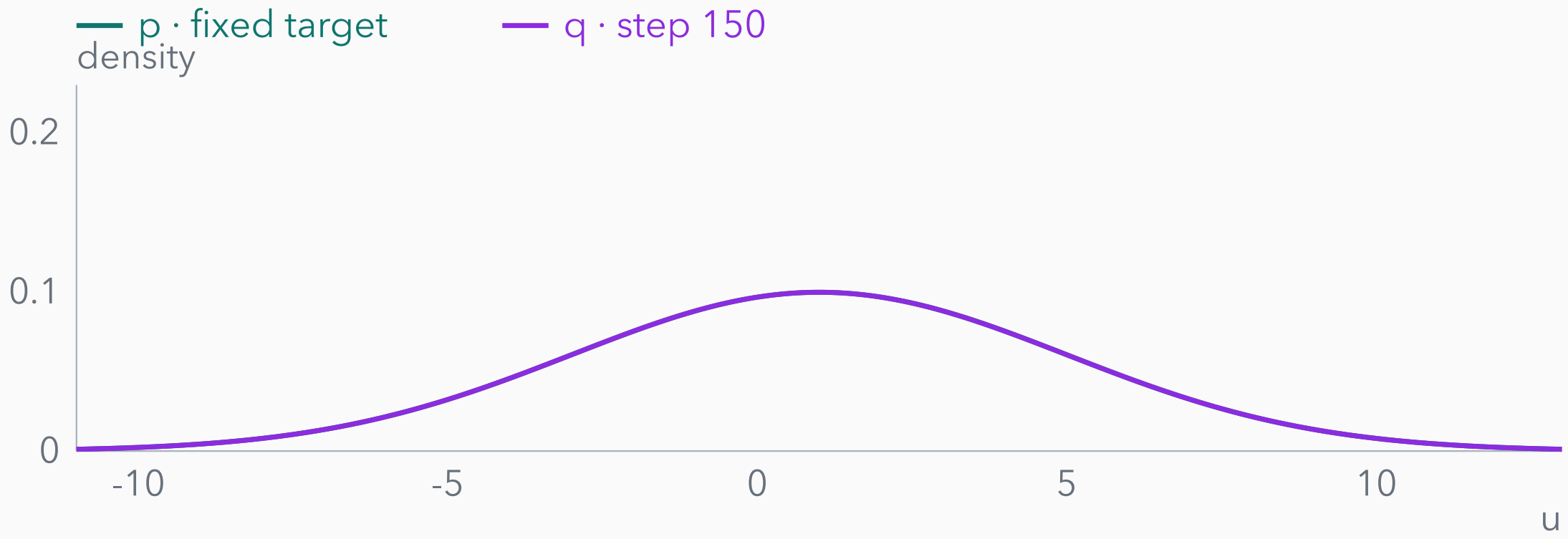
Rebuild, score, differentiate, update

```
q = Normal(mu, rho.exp())
loss = kl_divergence(q, p)
optimiser.zero_grad()
loss.backward()
optimiser.step()
```

Repeat with the updated parameters.

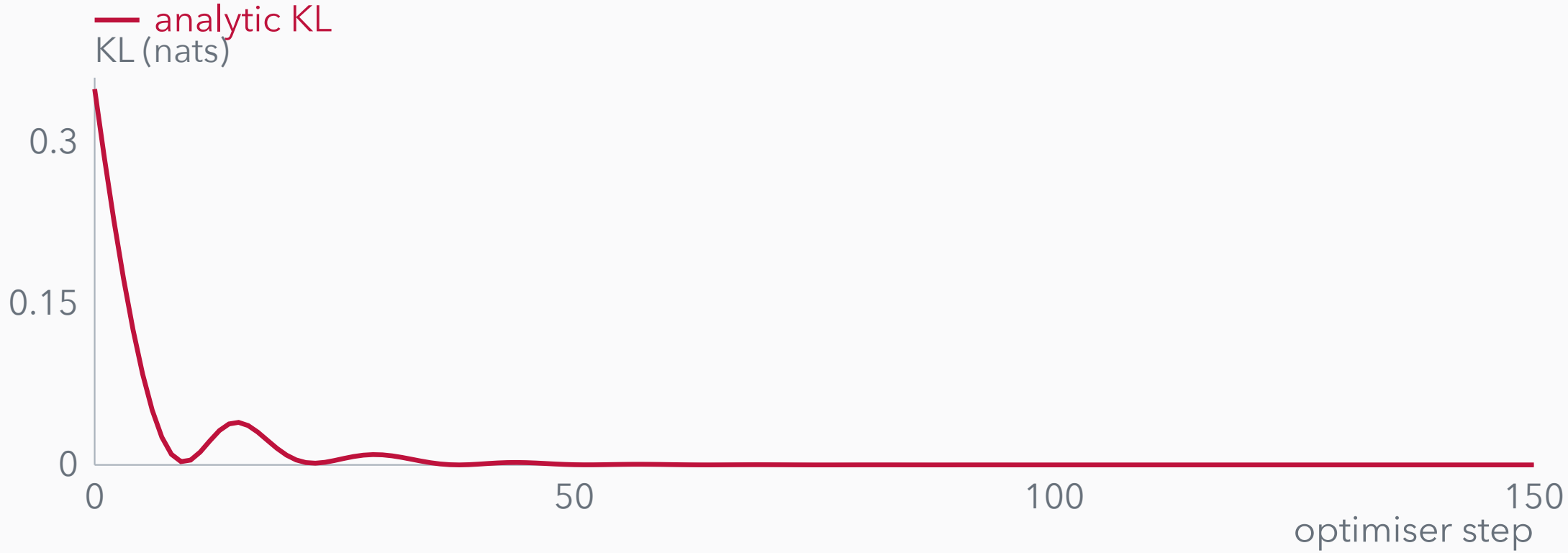
> Why clear gradients before backward?

Letting both parameters move removes the restriction



How close is the final measured fit?

A loss trace answers a specific question



This trace evaluates the exact Normal formula. It is not a Monte Carlo estimate.

> Would the same smooth trace be guaranteed for a sampled objective?

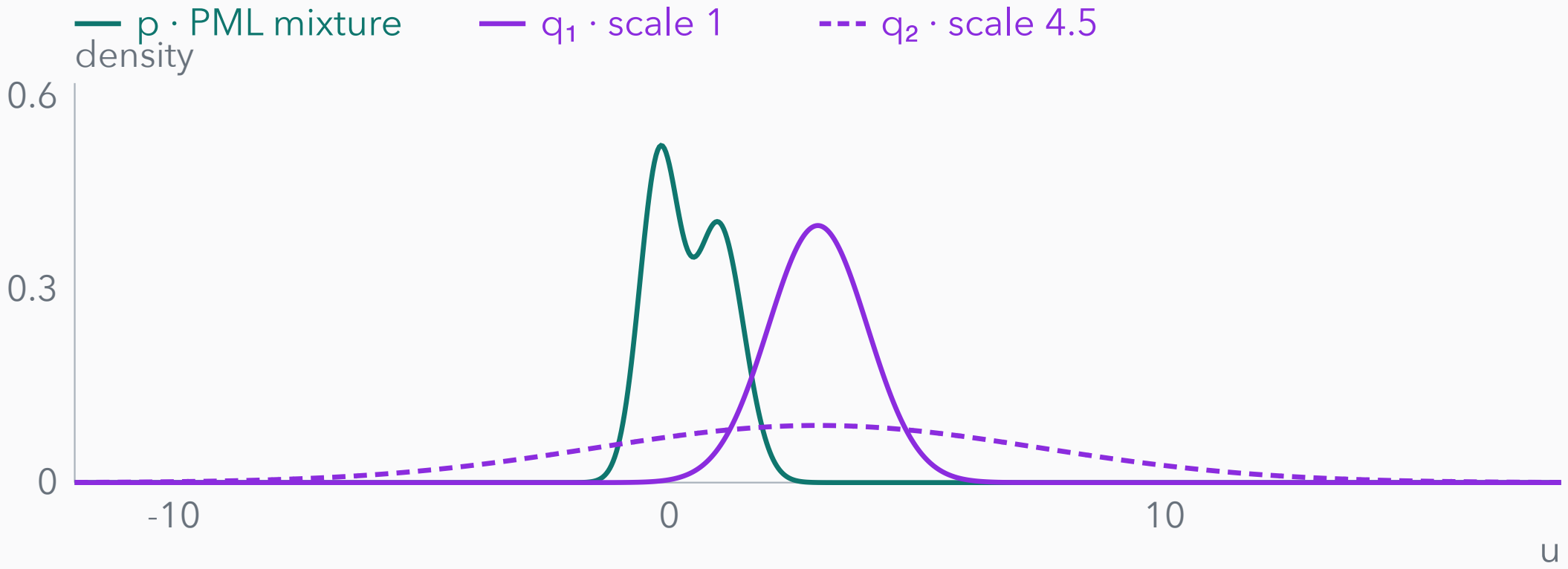
Replace the target; keep the definition

$$p(u) = \tfrac{1}{2}\mathcal{N}(u; -0.2, 0.4^2) + \tfrac{1}{2}\mathcal{N}(u; 1, 0.5^2)$$

We can evaluate this density at a proposed u.

> Does evaluating p(u) require a closed-form KL?

Which candidate wastes more mass far from the mixture?



Both candidates have mean 3. Their standard deviations are 1 and 4.5.

> Which candidate draws more often from far-away regions?

A missing library formula is not an undefined quantity

```
kl_divergence(q, mixture)
```

```
NotImplementedError:
KL not implemented for this distribution pair
```

What does this error establish, and what does it not establish?

Draw from q, score both densities, average

$$u_1, \ldots, u_S \sim q$$

$$\ell_i = \log q(u_i) - \log p(u_i)$$

$$\widehat{\mathrm{KL}}_S = \frac{1}{S} \sum_{i=1}^{S} \ell_i$$

Which dimension is averaged?

Validate the estimator before using the unfamiliar target

```
u = q.rsample((S,))
log_q = q.log_prob(u)
log_p = p.log_prob(u)
estimate = (log_q-log_p).mean()
```

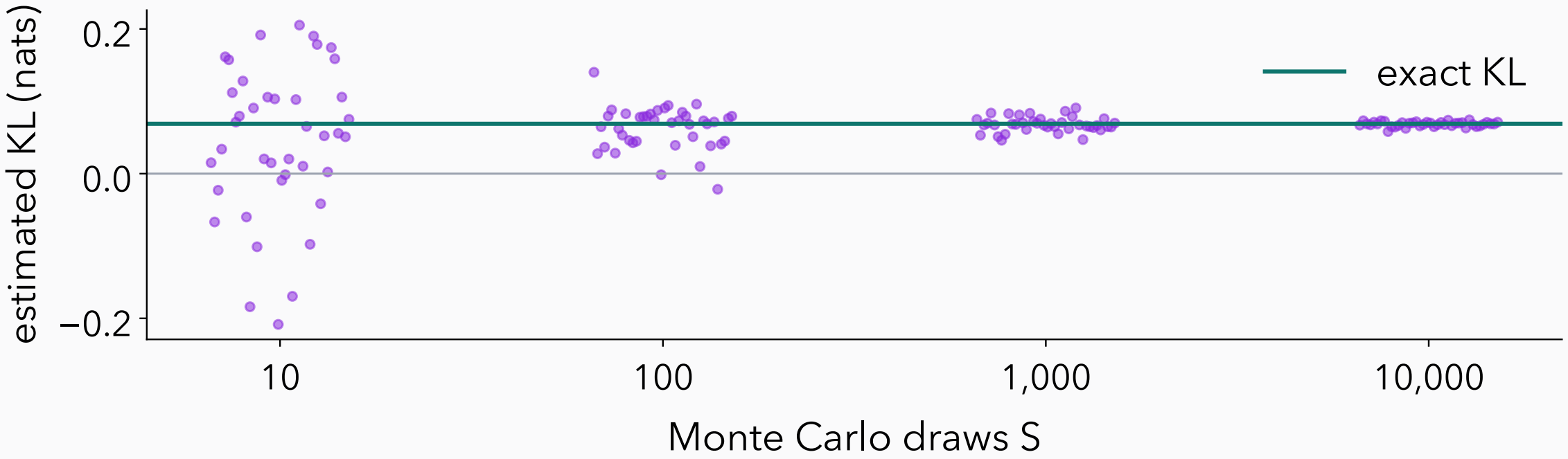
$$q = \mathcal{N}(1, 3^2), \quad p = \mathcal{N}(1, 4^2)$$

Exact KL	One MC estimate
0.0689	0.0671

S=10,000 · seed 1000

Should a finite estimate equal the exact answer digit for digit?

More draws reduce variability, not every individual error



Same p and q · 40 seeds per sample count · exact KL is the horizontal reference

A new seed changes the estimate. It does not change the exact KL.

> Must every larger-S estimate be closer than every smaller-S estimate?

Measured estimates · seeds1000–1039 · S=10,100,1000,10000 · float64

Five draws can give a negative KL estimate

u_i	$\log q(u_i) - \log p(u_i)$
5.623	−0.232
0.120	0.269
−5.536	−0.751
2.705	0.217
−2.254	0.030

$$\widehat{\mathrm{KL}}_5 \approx -0.0933, \qquad \mathrm{KL}(q\|p) \approx 0.0689$$

> Does this contradict nonnegative exact KL?

The estimated score separates these candidates clearly

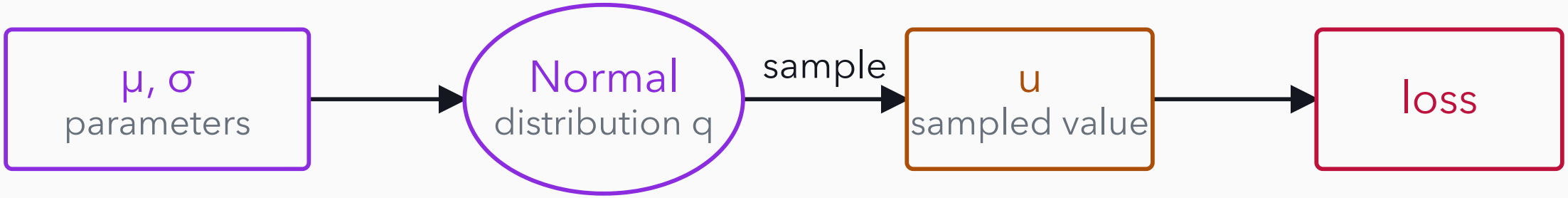
Candidate	MC estimate ± one SE	Numerical integral
q1: σ=1	9.545 ± 0.026	9.489
q2: σ=4.5	45.023 ± 0.212	45.152

q_1 has the smaller directional KL. The estimate is not an analytic formula.

> What role does the numerical integral play?

100,000 q draws · seeds402/403 · separate wide-domain numerical integration

A random draw is not automatically a differentiable path




```
u = Normal(mu, sigma).sample()
u.requires_grad   # False
```

Does this sample retain that path in PyTorch?

Ask the sample itself for a gradient

```
u = Normal(mu, sigma).sample()
u.backward()
```

RuntimeError: this tensor does not require grad.

What has this failure shown?

A backward call can run and still miss the intended path

```
u = q.sample((S,))
loss = (q.log_prob(u) - p.log_prob(u)).mean()
loss.backward()
```

log q still contains the parameters explicitly. The missing sample path has not been repaired.

> Would a nonempty gradient tensor prove this is the desired KL derivative?

Check the derivative, not just the existence of a tensor

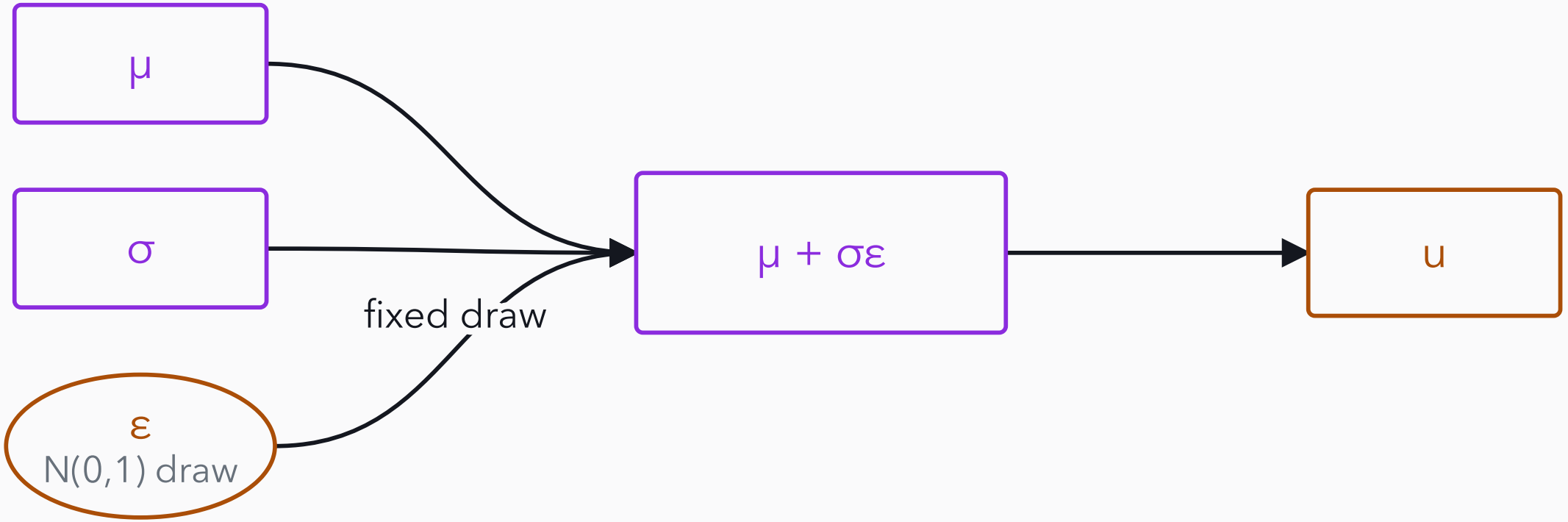
Derivative route	∂L/∂μ	∂L/∂σ
Naive detached samples	0.0013	−0.0003
Pathwise samples	−0.0622	−0.3752
Analytic KL	−0.0625	−0.3750

Same 50,000 chosen noise values · seed13 · q=Normal(0,2), p=Normal(1,4)

> Is Monte Carlo agreement expected to be exact?

Computed teaching example · natural logarithms · KL in nats

Keep the noise separate from the parameters



$$\epsilon \sim \mathcal{N}(0,1), \qquad u = \mu + \sigma\epsilon$$

For a fixed ε, which part is now deterministic?

What happens to this same draw when a parameter changes?

$$\epsilon = -0.75, \quad \mu = 0, \quad \sigma = 2 \quad \Longrightarrow \quad u = -1.5$$

$$\frac{\partial u}{\partial \mu} = 1$$

$$\frac{\partial u}{\partial \sigma} = \epsilon = -0.75$$

> Why can increasing σ move this particular u left?

Chosen scalar input ε=−0.75 · arithmetic and autograd checked

The loss gradient follows the same draw

$$L = (u-1)^2, \qquad \frac{\partial L}{\partial u} = 2(u-1) = -5$$

$$\frac{\partial L}{\partial \mu} = -5, \qquad \frac{\partial L}{\partial \sigma} = (-5)(-0.75) = 3.75$$

How does that derivative reach μ and σ?

A fixed-noise finite difference checks the path

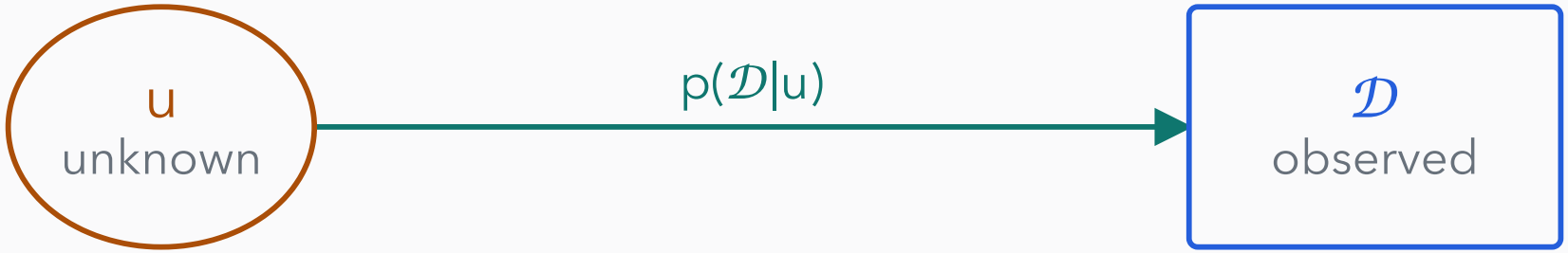
```
q = Normal(mu, sigma)
u = q.rsample((S,))
loss = (q.log_prob(u) - p.log_prob(u)).mean()
```

Derivative	Autograd	Finite difference
μ	−0.062466	−0.062466
σ	−0.372223	−0.372223

> What must stay fixed on both sides of the finite difference?

Measured check · seed88 · S=2048 · finite-difference $h=10^{-5}$

Now the target is an unknown posterior



Inference asks for $p(u \mid \mathcal{D})$

> Which object is our new target distribution?

Choose a family we can draw from and evaluate

$$q_\psi(u)$$

$$u \sim q_\psi, \qquad \log q_\psi(u)$$

> What operations do we need from this chosen family?

The direct score asks for the density we do not have

$$\mathrm{KL}(q_\psi(u)\|p(u \mid \mathcal{D})) = \mathbb{E}_{u\sim q_\psi}[\log q_\psi(u) - \log p(u \mid \mathcal{D})]$$

The unavailable term is $\log p(u \mid \mathcal{D})$.

> Which piece blocks direct evaluation of this score?

Replace the posterior with joint divided by evidence

$$p(u \mid \mathcal{D}) = \frac{p(\mathcal{D}, u)}{p(\mathcal{D})}$$

$$p(\mathcal{D}, u) = p(\mathcal{D} \mid u)\, p(u)$$

> Which denominator may still be difficult?

The log quotient becomes a difference

$$\log p(u \mid \mathcal{D}) = \log p(\mathcal{D}, u) - \log p(\mathcal{D})$$

$$-\log p(u \mid \mathcal{D}) = -\log p(\mathcal{D}, u) + \log p(\mathcal{D})$$

Why does evidence become positive when the posterior term is subtracted?

Keep the KL; change only the posterior term

$$\mathrm{KL}(q_\psi(u) \| p(u \mid \mathcal{D})) = \mathbb{E}_{u \sim q_\psi}[\log q_\psi(u) - \log p(u \mid \mathcal{D})]$$

$$= \mathbb{E}_{u \sim q_\psi}[\log q_\psi(u) - \log p(\mathcal{D}, u) + \log p(\mathcal{D})]$$

Which three terms now appear inside the average?

Constant with respect to what?

$$\mathbb{E}_{u\sim q_\psi}\left[\log q_\psi(u) - \log p(\mathcal{D}, u) + \log p(\mathcal{D})\right]$$

The model and observed data are fixed. Evidence depends on neither u nor ψ.

$$= \mathbb{E}_{u\sim q_\psi}\left[\log q_\psi(u) - \log p(\mathcal{D}, u)\right] + \log p(\mathcal{D})$$

> What justifies moving it outside?

Swap sides and reverse the bracket's signs

$$\mathrm{KL}(q_\psi(u)\|p(u \mid \mathcal{D})) = \mathbb{E}_{u\sim q_\psi}[\log q_\psi(u) - \log p(\mathcal{D}, u)] + \log p(\mathcal{D})$$

$$\log p(\mathcal{D}) = \mathbb{E}_{u\sim q_\psi}[\log p(\mathcal{D}, u) - \log q_\psi(u)] + \mathrm{KL}(q_\psi(u)\|p(u \mid \mathcal{D}))$$

Which term remains as a nonnegative gap?

A trainable objective emerges from the rearrangement

$$\mathrm{ELBO}(q_\psi) = \mathbb{E}_{u \sim q_\psi}\left[\log p(\mathcal{D}, u) - \log q_\psi(u)\right]$$

Evaluate log joint and log q at q's draws. Average to estimate the expectation.

> What do we still need to estimate in a general problem?

One fixed total, two parts

$$\log p(\mathcal{D}) = \mathrm{ELBO}(q_\psi) + \underbrace{\mathrm{KL}(q_\psi(u) \| p(u \mid \mathcal{D}))}_{\geq 0}$$

$$\mathrm{ELBO}(q_\psi) \leq \log p(\mathcal{D})$$

Raise the ELBO; lower the posterior KL by the same amount.

> Which optimisation direction improves q?

Optimise the ELBO without computing the evidence

$$\max_{\psi} \mathrm{ELBO}(q_\psi) \quad \Longleftrightarrow \quad \min_{\psi} [-\mathrm{ELBO}(q_\psi)]$$

Need: draws from q, log q, log joint.

Do not need: the normalised posterior or its evidence.

> What exactly makes it safe to omit from ψ's optimisation?

Return to the same coin and the same data

H H T H T H T H H H T T H T H

H T H T T H H H T T H T H T H

$$p(r) = \mathrm{Beta}(5, 3)$$

Fit q using the joint. Keep the known posterior for validation.

> Why retain a known answer if the optimiser does not use it?

The conjugate posterior is a reference answer

$$p(r \mid \mathcal{D}) = \mathrm{Beta}(5 + 17,\ 3 + 13)$$

$$= \mathrm{Beta}(22, 16)$$

> Does the optimiser need to call this posterior density?

Keep the prior’s normalising constant

$$p(r) = \frac{r^4(1-r)^2}{B(5,3)}$$

$$\log p(r) = 4\log r + 2\log(1-r) - \log B(5,3)$$

> Can we drop this constant when checking absolute ELBO values?

Each observed toss contributes one log probability

$$\log p(\mathcal{D} \mid r) = 17 \log r + 13 \log(1 - r)$$

One ordered sequence. There is no binomial coefficient in this convention.

> Why is there no factor counting different orders?

The joint is available at any proposed r

$$\log p(\mathcal{D}, r) = \log p(\mathcal{D} \mid r) + \log p(r)$$

```
log_likelihood = 17*log(r) + 13*log1p(-r)
log_joint = prior.log_prob(r) + log_likelihood
```

Where is the unknown evidence in this code?

Fit positive Beta shapes through unconstrained parameters

$$q_\psi(r) = \mathrm{Beta}(a, b)$$

$$a = \exp(\psi_a), \qquad b = \exp(\psi_b)$$

> Which quantities does the optimiser store and update?

One mean over draws; the data are already summed

```
a, b = psi.exp()
q = Beta(a, b)
r = q.rsample((512,))
terms = log_joint(r) - q.log_prob(r)
loss = -terms.mean()
```

r and terms each have shape [512]. Mean over Monte Carlo draws.

> Why is there no additional sum over the 512 draws before mean?

Inspect one actual update

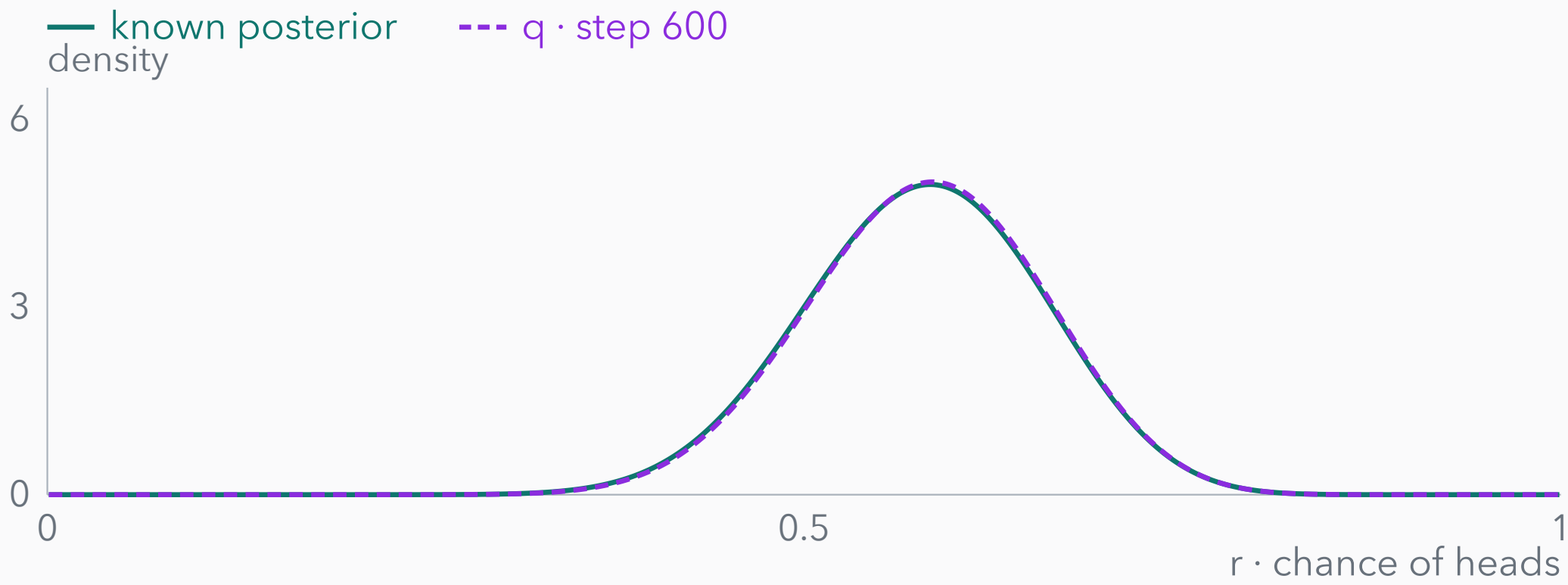
Quantity	Before	After one Adam step
a	10.0000	10.2532
b	10.0000	9.7531

$$\nabla_{\psi}[-\widehat{\mathrm{ELBO}}] \approx (-3.338,\ 2.960)$$

> Are the printed gradients with respect to a,b or ψ?

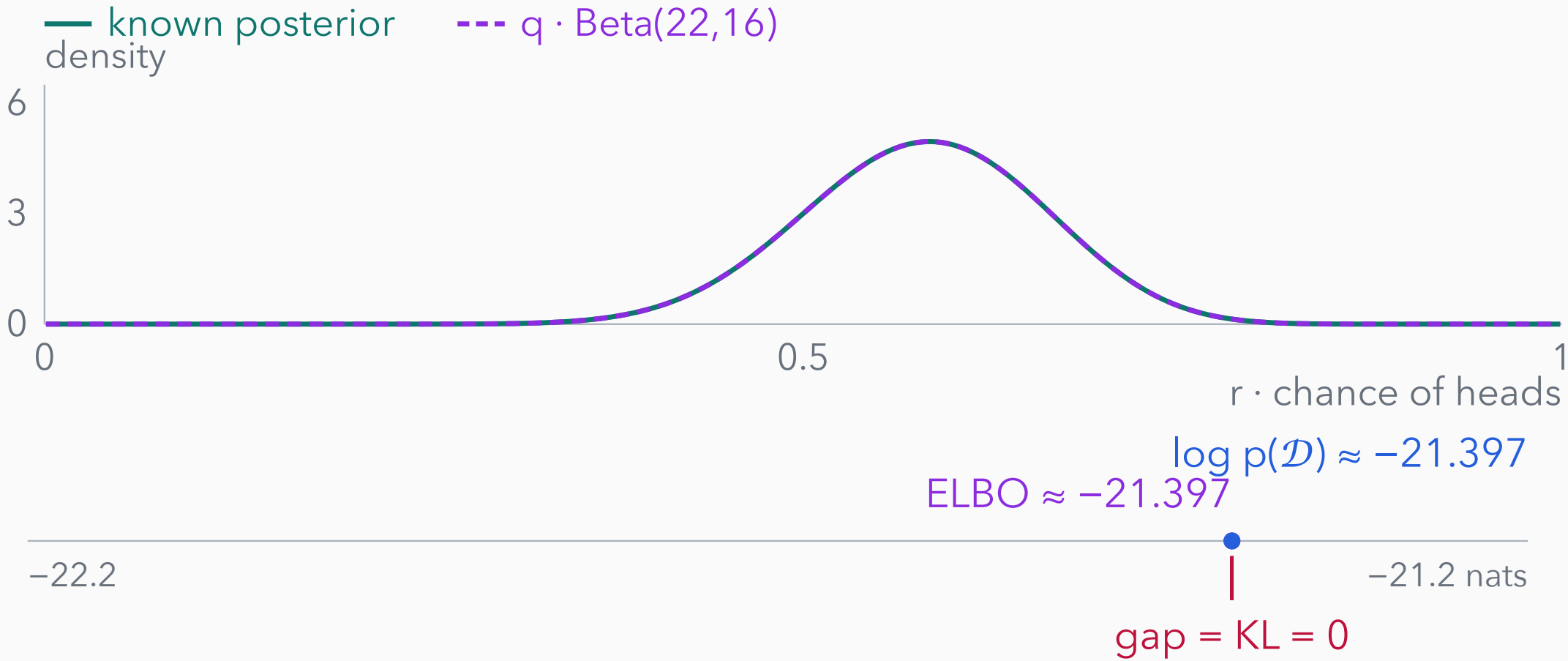
Measured first update · seed2401 · Adam lr=0.025 · 512 Beta draws

A measured fit using only the joint



Is the final q exactly the known posterior?

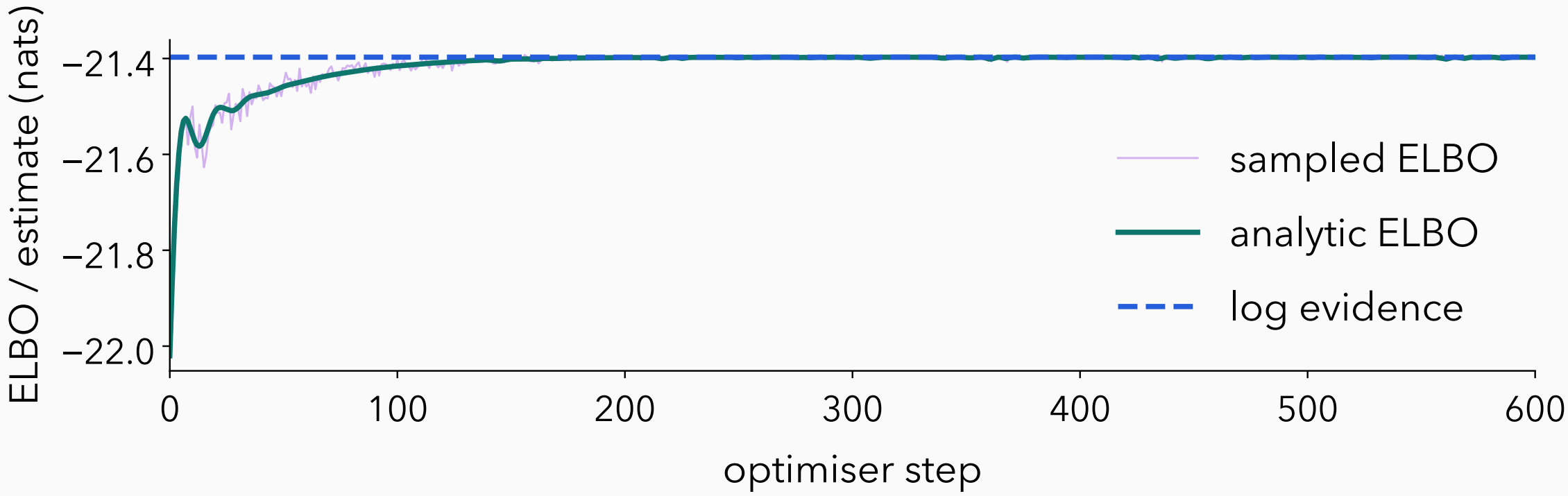
Check the fixed total with three selected candidates



What happens at the exact posterior?

Direct analytic Beta ELBO and independent posterior KL · selected candidates, not a training trace

An estimate is not itself guaranteed to be a lower bound



Same pre-update q at each step · fresh training draws · known log evidence reference

162 of 600 sampled estimates lie above the known evidence in this run.

> Which object remains a mathematical lower bound?

A higher or stable ELBO does not prove a zero gap

$$\mathrm{ELBO}(q_{\mathrm{new}}) > \mathrm{ELBO}(q_{\mathrm{old}})$$

The family may miss the posterior. Optimisation may stop short.

> Does a stable loss establish q=p(u|$\mathcal{D}$)?

For KL(q||p), where do the samples come from?

$$\mathrm{KL}(q\|p) = \boxed{\quad ? \quad}[\log q(u) - \log p(u)]$$

$$\boxed{\mathbb{E}_{u\sim q}}$$

> Why does sampling from p change the question?

What changes during fixed-model variational inference?

Object	Changes during this inference fit?
Observed data $\mathcal{D}$	?
Exact posterior $p(u\|\mathcal{D})$	?
Variational parameters ψ	?
A newly drawn u	?

Update ψ. Draw fresh u. The fixed data, model and exact posterior do not change.

> Is a new draw an optimiser parameter update?

Why can we omit the evidence from ψ's optimisation?

$$\mathrm{KL}(q_\psi \| p(u \mid \mathcal{D})) = -\mathrm{ELBO}(q_\psi) + \log p(\mathcal{D})$$

$$\nabla_\psi \log p(\mathcal{D}) = 0$$

> Constant with respect to which parameters?

A sensor has an unknown constant bias

$$u \sim \mathcal{N}(0, 1), \qquad y_i \mid u \sim \mathcal{N}(u, 0.5^2)$$

$$p(\mathcal{D}, u) = p(u) \prod_i p(y_i \mid u)$$

$$\max_{\psi} \mathbb{E}_{q_\psi(u)}[\log p(\mathcal{D}, u) - \log q_\psi(u)]$$

What objective would you implement?

Generative models will create this inference problem again

A hidden description can generate an observation.

Next: learn representations for images, then ask whether they can generate new images.

> What question should we ask about a good reconstruction?

Where does the complete Gaussian KL formula come from?

$$\log \frac{q(u)}{p(u)} = \log \frac{s}{\sigma} + \frac{(u-m)^2}{2s^2} - \frac{(u-\mu)^2}{2\sigma^2}$$

$$\mathbb{E}_q[(u-m)^2] = \sigma^2 + (\mu - m)^2, \quad \mathbb{E}_q[(u-\mu)^2] = \sigma^2$$

$$\mathrm{KL}(q\|p) = \log \frac{s}{\sigma} + \frac{\sigma^2 + (\mu-m)^2}{2s^2} - \frac{1}{2}$$

Where does the final −1/2 come from?

Why is exact KL nonnegative?

$$\mathrm{KL}(q\|p) = -\mathbb{E}_q\left[\log \frac{p(u)}{q(u)}\right]$$

$$\geq -\log \mathbb{E}_q\left[\frac{p(u)}{q(u)}\right]$$

$$= -\log 1 = 0$$

What makes the final expectation one?

Pathwise gradients are one route, not the only route

$$\nabla_\psi \mathbb{E}_{q_\psi}[f_\psi(u)] = \mathbb{E}_{q_\psi}[f_\psi(u)\nabla_\psi \log q_\psi(u) + \nabla_\psi f_\psi(u)]$$

A score-function estimator accounts explicitly for the changing sampling distribution.

> Does this identity require u=μ+σε?

Differentiation-under-the-integral identity · regularity assumptions apply

A mean must say which dimension it averages

Object	Shape	Correct reduction
S scalar Beta draws	[S]	mean over S
S d-dimensional joint log densities	[S]	mean over S
S factorised coordinate log densities	[S,d]	sum event d, then mean S

MultivariateNormal.log_prob already sums the event. Do not sum the sample axis a second time.

> What goes wrong if you sum S before subtracting log q from the joint?

An ordered sequence and a count have different evidence

$$p(H = 17 \mid r) = \binom{30}{17} r^{17}(1 - r)^{13}$$

Observed object	Log evidence
One ordered sequence	−21.39724
Only the head count	−2.79624

> Does this constant change the posterior or its KL gap?

Exact convention comparison · log binomial coefficient computed with log-gamma

Check the sampled objective with analytic Beta moments

$$\mathbb{E}_q[\log r] = \mathrm{digamma}(a) - \mathrm{digamma}(a+b)$$

$$\mathbb{E}_q[\log(1-r)] = \mathrm{digamma}(b) - \mathrm{digamma}(a+b)$$

Insert these into expected log likelihood + expected log prior − expected log q.

> How was the identity checked non-circularly?